

Proves

R4 Cheng

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Cost function

L_2 :

$$J(\vec{w}) = \frac{1}{2} \sum_{(\vec{x}, y) \in D} (f_{\vec{w}}(\vec{x}) - y)^2 + \lambda \|\vec{w}\|_2^2$$

> Euclidean norm: $\|\vec{w}\|_2 = \sqrt{w_0^2 + w_1^2 + \dots + w_n^2}$

L_1 :

$$J(\vec{w}) = \frac{1}{2} \sum_{(\vec{x}, y) \in D} (f_{\vec{w}}(\vec{x}) - y)^2 + \lambda \|\vec{w}\|_1^2$$

> Manhattan norm: $\|\vec{w}\|_1 = |w_0| + |w_1| + \dots + |w_n|$

Target:

$$\min_{\vec{w}} J(\vec{w})$$

(Batch) Gradient Descent

Start from an arbitrary initial weight vector $\vec{w}$.

$$\vec{w}' \leftarrow \vec{w} - \alpha \Delta J(\vec{w}) = \vec{w} - \alpha \frac{\partial}{\partial \vec{w}} J(\vec{w})$$

> α is called learning rate

And for a single training example $(\vec{x}, y)$, we have:

$$\vec{w}' \leftarrow \vec{w} - \alpha (\vec{w} \cdot \vec{x} - y) \vec{x} = \vec{w} + \alpha (y - \vec{w} \cdot \vec{x}) \vec{x}$$

Aka. the least mean square (LMS) update rule.

> vs perceptron: $\vec{w}' = \vec{w} + y \cdot \vec{x}$

$$\begin{aligned} \text{Derivation : } \frac{\partial}{\partial \vec{w}} J(\vec{w}) &= \frac{\partial}{\partial \vec{w}} \frac{1}{2} (f(\vec{x}) - y)^2 \\ &= 2 \cdot \frac{1}{2} (f(\vec{x}) - y) \cdot \frac{\partial}{\partial \vec{w}} (f(\vec{x}) - y) \quad // \text{ Chain Rule} \\ &= (\vec{w} \cdot \vec{x} - y) \cdot \frac{\partial}{\partial \vec{w}} (\vec{w} \cdot \vec{x} - y) \\ &= (\vec{w} \cdot \vec{x} - y) \vec{x} \end{aligned}$$

Therefore, for the whole dataset D :

$$\vec{w}' \leftarrow \vec{w} + \alpha \sum_{(\vec{x}, y) \in D} (y - \vec{w} \cdot \vec{x}) \vec{x}$$

Stochastic Gradient Descent

Streamline BGD:

$$\vec{w}' \leftarrow \vec{w} + \alpha (y_i - \vec{w} \cdot \vec{x}_i) \vec{x}_i$$

Where $(\vec{x}_i, y_i)$ is a randomly picked training example from D .

Mini-Batch Gradient Descent

> Balance between GD and SGD

$$\vec{w}' \leftarrow \vec{w} + \alpha \sum_{(\vec{x}, y) \in D_m} (y - \vec{w} \cdot \vec{x}) \vec{x}$$

Where $D_m \subset D$ is a randomly picked m examples of training data.

Ridge regression gradient descent

> L_2 , because it is differentiable

$$\vec{w}' \leftarrow \vec{w} + \alpha \sum_{(\vec{x}, y) \in D} (y - \vec{w} \cdot \vec{x}) \vec{x} - \lambda \vec{w}$$

The function of $\lambda \vec{w}$ is to decrease the magnitude of $\vec{w}$, leading to reduced model complexity and overfitting.